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Q1. Food Delivery Billing System (Multiple Inheritance using Two Interfaces)

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.*;
interface DeliveryCharge{
    double calculateDeliveryCharge(double cost);
}
interface CouponDiscount{
    double calculateDiscount(double cost,String coupon);
}
class FoodOrder implements DeliveryCharge,CouponDiscount{
    public double calculateDeliveryCharge(double cost){
        return (cost>=500)?0:50;
    }
    public double calculateDiscount(double cost,String coupon){
        return coupon.equals("SAVE10"?cost*0.10:0;

    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        double cost=sc.nextDouble();
        sc.nextLine();
        String coupon=sc.nextLine();
        FoodOrder f=new FoodOrder();
        double delivery=f.calculateDeliveryCharge(cost);
        double discount=f.calculateDiscount(cost,coupon);
        double bill=cost+delivery-discount;
        System.out.println("Food Cost : "+cost);
        System.out.println("Delivery Charge : "+delivery);
        System.out.println("Discount : "+discount);
        System.out.println("Final Bill : "+bill);
    }
}
```

INPUT

600
SAVE10

OUTPUT

Food Cost : 600.0
Delivery Charge : 0.0
Discount : 60.0
Final Bill : 540.0

Q2. Hotel Billing System (Multiple Inheritance using One Class and One Interface)

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
class Main
{
    public static void main(String[]args)
    {
        Scanner in=new Scanner(System.in);
        HotelBill h=new HotelBill();
        h.Room_number=in.nextInt();
        in.nextLine();
        h.Name=in.nextLine();
        h.Rent=in.nextDouble();
        h.calculateServiceCharge();
        h.finalbill();
        h.display();
    }
}
class Room
{
    public int Room_number;
    public String Name;
    public double Rent;
}
interface Service_charge
{
    public double calculateServiceCharge();
}
class HotelBill extends Room implements Service_charge
{
    public double final_bill;
    public double ServiceCharge;
    public double calculateServiceCharge()
    {
        if(Rent>=5000)
        {
            ServiceCharge=Rent*0.12;
        }
        else
        {
            ServiceCharge=Rent*0.08;
        }
        return ServiceCharge;
    }
    public void finalbill()
    {
        final_bill=ServiceCharge+Rent;
    }
    void display()
    {
        System.out.println("Room Number : "+Room_number);
        System.out.println("Guest Name : "+Name);
        System.out.println("Room Rent : "+Rent);
        System.out.println("Service Charge : "+ServiceCharge);
        System.out.println("Final Bill : "+final_bill);
    }
}
```

INPUT

```
205
Ananya
6000
```

Report

OUTPUT

```
Room Number : 205
Guest Name : Ananya
Room Rent : 6000.0
Service Charge : 720.0
Final Bill : 6720.0
```